



Produced-Water Lithium Extraction

TBAC/DA DES + TOPO from Smackover feed basis to staged extraction and screening TEA

Chemical engineering technical review
Smackover Li/Na feed basis → pretreatment → TBAC/DA DES + TOPO → ALAMO surrogate → PrOMMiS/IDAES
May 15, 2026

2026-05-15

Produced-Water Lithium Extraction

Produced-Water Lithium Extraction
TBAC/DA DES + TOPO from Smackover feed basis to staged extraction and screening TEA

Chemical engineering technical review
Smackover Li/Na feed basis → pretreatment → TBAC/DA DES + TOPO → ALAMO surrogate → PrOMMiS/IDAES
May 15, 2026

I am going to walk through a produced-water lithium case study as an integration package, not as a broad market overview. The chain is deliberately narrow: start with a Smackover feed, define what pretreatment removes, map the Rezaee TBAC/DA DES plus TOPO Li/Na response into compact transfer terms, route those terms through an ALAMO surrogate, and then carry them into PrOMMiS/IDAES staged extraction and screening TEA. The reason to keep the chain this explicit is that each layer has a different job. Feed chemistry defines the severity. The solvent response defines selectivity. The process model defines staged recovery. The economics tell us which improvements are worth chasing first.

A pretreated Smackover Li/Na stream connects realistic produced-water chemistry to staged extraction and screening-level economics.

2026-05-15

Produced-Water Lithium Extraction

A pretreated Smackover Li/Na stream connects realistic produced-water chemistry to staged extraction and screening-level economics.

The thesis is that the useful technical unit is not a raw brine and not a stand-alone extractant number. The useful unit is a pretreated Li/Na stream whose transfer behavior can be propagated through stages and costs. That is why the case study starts from produced-water chemistry, but it does not try to put every raw-brine component into the contactor. The divalent and oil-side burdens are handled at the boundary. What moves forward is the Li/Na transfer problem: how much lithium moves, how much sodium follows, and how those two numbers change the process result.

CASE BASIS

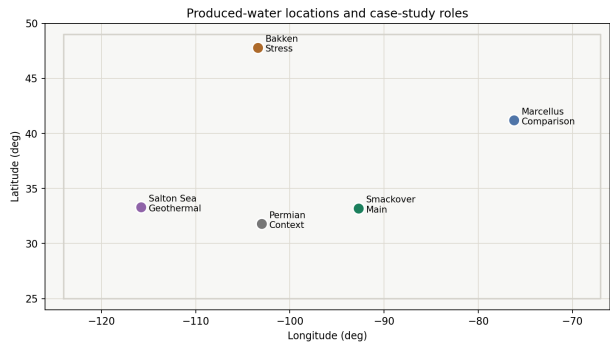
2026-05-15

Produced-Water Lithium Extraction
└ Case Basis

CASE BASIS

I am starting with the feed because that is the part of the story where produced-water realism matters most. Lithium extraction can look artificially clean if the feed basis is vague. Here, the feed basis is tied to a southern Arkansas Smackover row, and the rest of the section explains why that row is a useful main case rather than just another sensitivity point.

Smackover Is The Main Produced-Water Case



Case basis

Southern Arkansas Smackover combines meaningful lithium grade with high-salinity major-ion chemistry.

- MS-2 feed: 168 mg/L Li, 64 100 mg/L Na, and 305 000 mg/L TDS.
- High-observed Smackover sensitivity: 252 mg/L Li.
- Marcellus, Bakken, Permian, and Salton Sea remain comparison cases.

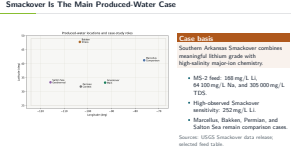
Sources: USGS Smackover data release; selected feed table.

2026-05-15

Produced-Water Lithium Extraction

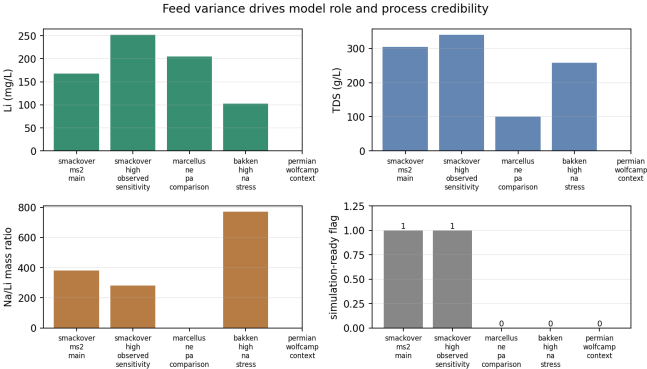
└ Case Basis

└ Smackover Is The Main Produced-Water Case



The map is here to anchor the case geographically before we start talking about extraction numbers. Smackover is not being treated as the only possible produced-water lithium source. It is being treated as the strongest main case for this package because it combines meaningful lithium grade with very severe salinity. The MS-2 feed is the base row: 168 milligrams per liter lithium, 64,100 milligrams per liter sodium, and 305,000 milligrams per liter total dissolved solids. The high-observed Smackover row is kept as a feed-grade sensitivity. That separation is important because it lets the main case stay traceable while still showing what better lithium grade does to the process.

Feed Variance Sets The Model Role



- Smackover MS-2 anchors the main flowsheet and TEA case.
- The high-Li Smackover row tests feed-grade sensitivity.
- Incomplete major-ion rows stay outside the main process calculation.

Feed	Role
MS-2	main case
High-Li Smackover	sensitivity
Marcellus/Bakken	context

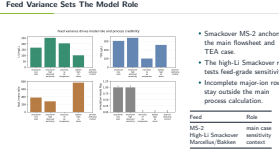
2026-05-15

Produced-Water Lithium Extraction

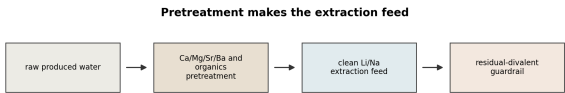
- └ Case Basis

- └ Feed Variance Sets The Model Role

This matrix explains why I do not treat every brine row the same way. MS-2 is the main calculation basis because it has the complete chemistry needed to carry a process case. The high-lithium Smackover row is a targeted sensitivity because it answers a specific question: how much does feed grade move the economics if the same extraction structure is used? The other basins are useful context, but they are not allowed to dilute the main calculation. That matters because feed variance is real, and a produced-water model should show that variance without pretending that every comparison row is equally ready for flowsheet use.



Pretreatment Creates The Li/Na Extraction Feed



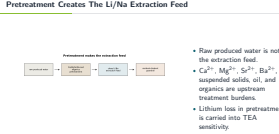
- Raw produced water is not the extraction feed.
- Ca^{2+} , Mg^{2+} , Sr^{2+} , Ba^{2+} , suspended solids, oil, and organics are upstream treatment burdens.
- Lithium loss in pretreatment is carried into TEA sensitivity.

2026-05-15

Produced-Water Lithium Extraction

└ Case Basis

└ Pretreatment Creates The Li/Na Extraction Feed



The important boundary on this slide is that raw produced water is not the extraction feed. The DES contactor sees a Li/Na-focused stream after the obvious upstream burdens are removed or controlled: calcium, magnesium, strontium, barium, suspended solids, oil, and organics. That does not make pretreatment invisible. Lithium loss across pretreatment is carried into the TEA sensitivity, so the boundary still has an economic cost. The purpose is to avoid mixing two problems together. Pretreatment determines whether the brine can be made suitable for contact. The extraction model then asks how Li and Na transfer once that boundary has been crossed.

2026-05-15

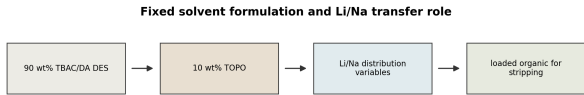
Produced-Water Lithium Extraction
└─ Solvent And Transfer Surface

SOLVENT AND TRANSFER SURFACE

SOLVENT AND TRANSFER SURFACE

Now I am moving from feed realism to the transfer surface. The question in this section is simple: once the pretreated Li/Na stream reaches the solvent, what variables describe the extraction response well enough to drive a staged process model?

TBAC/DA DES + TOPO Is The Active Solvent



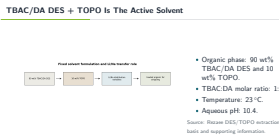
- Organic phase: 90 wt% TBAC/DA DES and 10 wt% TOPO.
- TBAC:DA molar ratio: 1:2.
- Temperature: 23 °C.
- Aqueous pH: 10.4.

Source: Rezaee DES/TOPO extraction basis and supporting information.

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

└ TBAC/DA DES + TOPO Is The Active Solvent



This is the solvent definition, and it should be read as a constraint on the whole case study. The organic phase is not a generic solvent stand-in; it is the Rezaee TBAC/DA deep eutectic solvent with 10 weight percent TOPO. The TBAC to decanoic acid molar ratio is one to two, the temperature basis is 23 degrees Celsius, and the aqueous pH basis is 10.4. Those values matter because selectivity is not only a property of lithium and sodium; it is a property of the feed, solvent chemistry, pH, temperature, and phase ratio together. This slide locks that chemistry before any surrogate or TEA number is introduced.

Stage-Response Surface Spans Source And Smackover Domains

Domain	Rows	Use
Rezaee DOE rows	26	Li/Na target surface
optimum rows	5	high-response anchors
real-brine row	1	high-Na/Li bound
process design	523	Smackover sensitivity

Retained source anchors	
optimized Li extraction	48.57%
optimized Li/Na selectivity	4.41
model-brine Li extraction	51.63%
model-brine Na extraction	9.97%

- The repaired package route evaluates 26 phase-tagged reactive rows.
- The surrogate uses 31 finite DOE and optimum target rows.
- Smackover MS-2 is outside the original Na/Li range and is marked as an extrapolated process case.

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

└ Stage-Response Surface Spans Source And Smackover Domains

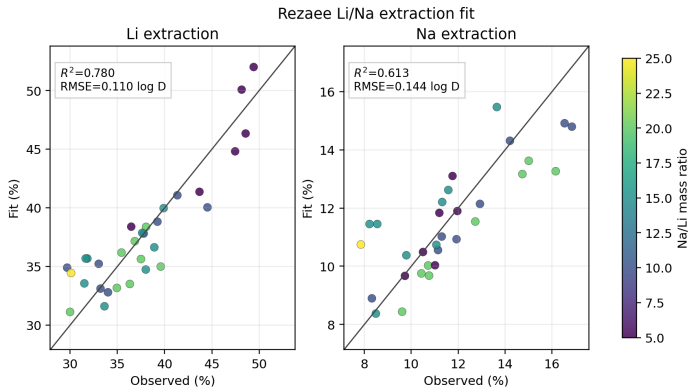
Stage-Response Surface Spans Source And Smackover Domains			
Domain	Rows	Use	
Rezaee DOE rows	26	Li/Na target surface	
optimum rows	5	high-response anchors	
real-brine row	1	high-Na/Li bound	
process design	523	Smackover sensitivity	

Retained source anchors	
optimized Li extraction	48.57%
optimized Li/Na selectivity	4.41
model-brine Li extraction	51.63%
model-brine Na extraction	9.97%

- The repaired package route evaluates 26 phase-tagged reactive rows.
- The surrogate uses 31 finite DOE and optimum target rows.
- Smackover MS-2 is outside the original Na/Li range and is marked as an extrapolated process case.

This table explains exactly what the current transfer surface is allowed to say. The Rezaee DOE rows give the main Li and Na extraction response. The five optimum rows keep the surface tied to the higher-response region near the reported 48.57 percent lithium extraction and 4.41 Li over Na selectivity. The real-brine row is retained because it is the only direct high-sodium anchor. The repaired ePC-SAFT route now evaluates the 26 phase-tagged reactive rows, so the thermodynamic layer is active. The Smackover process rows still stretch the Na over Li ratio far beyond the original DOE range, so I read them as process sensitivities rather than as a claim that the paper fully measured that brine.

ALAMO-Ready Surrogate Carries The Current Li/Na Targets



- Inputs: temperature, pH, TOPO loading, and Na/Li ratio.
- Outputs: $\log D_{Li}$ and $\log D_{Na}$ for process transfer.
- Current target-row fit:
 $R^2 = 0.780$ for Li and
 $R^2 = 0.613$ for Na.

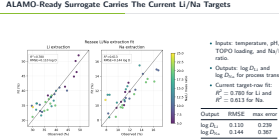
Output	RMSE	max error
$\log D_{Li}$	0.110	0.239
$\log D_{Na}$	0.144	0.367

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

└ ALAMO-Ready Surrogate Carries The Current Li/Na Targets

This plot is the current surrogate check after replacing the older transfer values with the Rezaee target rows from the repaired package evaluation. The surface is intentionally simple: temperature, pH, TOPO loading, and Na over Li ratio go in, and log distribution terms for lithium and sodium come out. Lithium fits more tightly than sodium, with an R^2 of 0.780 versus 0.613. That matters for how I describe the result. I would not call this a final thermodynamic closure. I would call it a current process surrogate that can carry source-traced Li and Na transfer into PrOMMiS and IDAES while the direct activity and fugacity route is connected.



Paper-Response Variables Feed The Process Model

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$
$$S_{Li/Na} = \frac{D_{Li}}{D_{Na}}$$



Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

Rezaee anchors, $O/A = 1$

Optimized 10 wt% TOPO: $E_{Li} = 48.57\%$, $S_{Li/Na} = 4.41$, so $E_{Na} = 11.01\%$.
Model-brine row: $E_{Li} = 51.63\%$ and $E_{Na} = 9.97\%$.

2026-05-15

Produced-Water Lithium Extraction

- └ Solvent And Transfer Surface

- └ Paper-Response Variables Feed The Process Model

Paper-Response Variables Feed The Process Model

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$
$$S_{Li/Na} = \frac{D_{Li}}{D_{Na}}$$

Rezaee anchors, $O/A = 1$

Optimized 10 wt% TOPO: $E_{Li} = 48.57\%$, $S_{Li/Na} = 4.41$, so $E_{Na} = 11.01\%$.
Model-brine row: $E_{Li} = 51.63\%$ and $E_{Na} = 9.97\%$.

This is where extraction percentages become process variables. For each species, the distribution coefficient is the extracted fraction divided by the raffinate fraction, corrected by the organic-to-aqueous ratio. Once D_{Li} and D_{Na} are defined, selectivity is just the ratio of those two coefficients. That conversion is important because staged extraction models do not need a paper table copied verbatim; they need transfer terms that can be applied stage by stage. The Rezaee anchor at $O/A = 1$ gives 48.57 percent lithium extraction and 11.01 percent sodium extraction for the optimized TOPO case, which is enough to define a process-ready Li/Na transfer pair.

2026-05-15

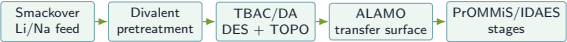
Produced-Water Lithium Extraction
└ Process And Economics

PROCESS AND ECONOMICS

PROCESS AND ECONOMICS

At this point the talk shifts from chemistry response to process consequence. The next section asks what the Li/Na transfer pair means after one stage, three stages, five stages, and then after the same recovery values are carried into a screening TEA.

PrOMMiS/IDAES Uses Active Li And Na Transfer Terms



Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

Stages	Li recovery	Na co-transfer
1	19.508%	7.065%
3	47.850%	19.734%
5	66.212%	30.675%

- The process layer uses PrOMMiS SolventExtraction objects with IDAES MSContactor units.
- Lithium and sodium material-transfer terms are active.
- Chloride transfer is held at zero inside the Li/Na extraction boundary.

2026-05-15

Produced-Water Lithium Extraction

- └ Process And Economics

- └ PrOMMiS/IDAES Uses Active Li And Na Transfer Terms

PrOMMiS/IDAES Uses Active Li And Na Transfer Terms

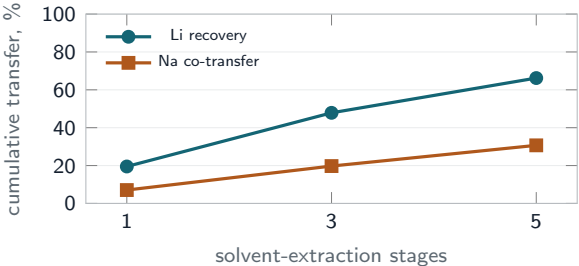
• The process layer uses PrOMMiS SolventExtraction objects with IDAES MSContactor units.

• Lithium and sodium material-transfer terms are active.

• Chloride transfer is held at zero inside the Li/Na extraction boundary.

This is the transfer from process variables to unit operations. The process layer uses PrOMMiS solvent-extraction objects with IDAES MSContactor units, and the active material-transfer terms are lithium and sodium. Chloride is held at zero transfer inside this Li/Na extraction boundary because the immediate question is selectivity between the target ion and the major co-transferred alkali ion. The table shows the effect of staging using the refreshed source-row surrogate. One stage recovers 19.508 percent lithium. Three stages recover 47.850 percent. Five stages recover 66.212 percent, but sodium co-transfer rises to 30.675 percent. That is the process tradeoff the economics has to price.

Three Stages Recover 47.85% Lithium



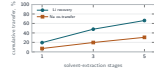
- Smackover MS-2 at $O/A = 1$ and 10 wt% TOPO reaches 47.85% cumulative lithium recovery after three stages.
- Sodium co-transfer is 19.73% at the same condition.
- Five stages raise lithium recovery to 66.21% with a larger sodium burden.

2026-05-15

Produced-Water Lithium Extraction

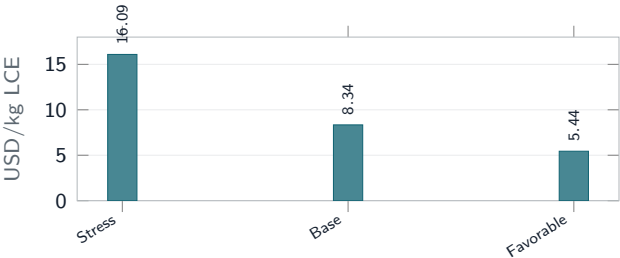
└ Process And Economics

└ Three Stages Recover 47.85% Lithium



- Smackover MS-2 at $O/A = 1$ and 10 wt% TOPO reaches 47.85% cumulative lithium recovery after three stages.
- Sodium co-transfer is 19.73% at the same condition.
- Five stages raise lithium recovery to 66.21% with a larger sodium burden.

Screening TEA Centers At 8.34 USD/kg LCE



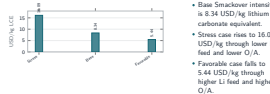
- Base Smackover intensity is 8.34 USD/kg lithium carbonate equivalent.
- Stress case rises to 16.09 USD/kg through lower Li feed and lower O/A.
- Favorable case falls to 5.44 USD/kg through higher Li feed and higher O/A.

2026-05-15

Produced-Water Lithium Extraction

Process And Economics

Screening TEA Centers At 8.34 USD/kg LCE



The TEA slide should be framed as prioritization. The base Smackover case is 8.34 dollars per kilogram lithium carbonate equivalent. The stress case moves to 16.09 dollars per kilogram because the lower lithium feed and lower organic-to-aqueous ratio reduce product mass. The favorable case moves to 5.44 dollars per kilogram because the high-lithium feed and higher phase ratio recover more product from the same nominal flow. The message is not that this is a final cost estimate. The message is that the model already ranks the dominant levers: feed grade, phase ratio, and staged recovery.

Economic Levers Are Feed Grade, Stages, And Losses

Scenario	Recovery	Cost	Product	Net
stress	25.04%	16.09	44.43	-0.18
base Smackover	47.85%	8.34	142.65	1.66
favorable	71.97%	5.44	383.13	11.33

Cost in USD/kg LCE; product in t/y; net in million USD/y.

Screening boundary

The TEA compares operating sensitivities for prioritization; it does not replace equipment design, residence-time sizing, solvent-loss experiments, or product-pricing analysis.

- Feed lithium grade is the strongest favorable lever.
- O/A ratio shifts recovery and solvent-contacting cost together.
- The stress case is not attractive at the assumed price basis.

2026-05-15

Produced-Water Lithium Extraction

└ Process And Economics

└ Economic Levers Are Feed Grade, Stages, And Losses

Scenario	Recovery	Cost	Product	Net	Screening boundary
stress	25.04%	16.09	44.43	-0.18	The TEA compares operating sensitivities for prioritization; it does not replace equipment design, residence-time sizing, solvent-loss experiments, or product-pricing analysis. • Feed lithium grade is the strongest favorable lever. • O/A ratio shifts recovery and solvent-contacting cost together. • The stress case is not attractive at the assumed price basis.
base Smackover	47.85%	8.34	142.65	1.66	
favorable	71.97%	5.44	383.13	11.33	

Cost in USD/kg LCE; product in t/y; net in million USD/y.

This table makes the current TEA sensitivity explicit. The stress case has only 25.04 percent lithium recovery, 44.43 tonnes per year of lithium carbonate equivalent, and a slightly negative net value at the assumed price. The base Smackover case reaches 47.85 percent recovery and 142.65 tonnes per year. The favorable case reaches 71.97 percent recovery and 383.13 tonnes per year. The ranking is clear: higher lithium feed and better transfer conditions move the economics more than small accounting choices.

THERMODYNAMIC EXTENSION

2026-05-15

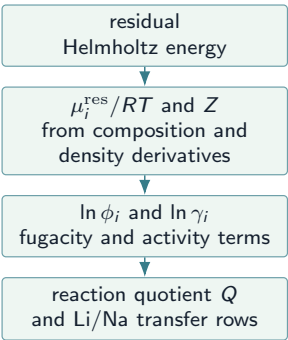
Produced-Water Lithium Extraction
└─ Thermodynamic Extension

THERMODYNAMIC EXTENSION

The final section shows where ePC-SAFT enters the case study. The downstream process structure is already shaped to receive activity and fugacity based transfer rows without rebuilding the entire presentation.

ePC-SAFT Supplies Activity And Fugacity Terms

$$\begin{aligned} \tilde{a}^{\text{res}} &= \tilde{a}^{\text{hc}} + \tilde{a}^{\text{disp}} + \tilde{a}^{\text{assoc}} + \tilde{a}^{\text{DH}} + \tilde{a}^{\text{Born}} \\ \tilde{a}^{\text{DH}} &= -\frac{\kappa e^2}{12\pi\epsilon_0\epsilon_r k_B T} \sum_i x_i z_i^2 \chi_i \\ \kappa &= \sqrt{\frac{\rho e^2}{k_B T \epsilon_0 \epsilon_r} \sum_j x_j z_j^2} \\ \chi_i &= \frac{3}{(\kappa d_i)^3} \left[\frac{3}{2} + \ln(1 + \kappa d_i) - 2(1 + \kappa d_i) + \frac{1}{2}(1 + \kappa d_i)^2 \right] \\ \tilde{a}^{\text{Born}} &= -\frac{e^2}{4\pi\epsilon_0 k_B T} \sum_i x_i z_i^2 \sum_{\delta \in \mathcal{D}} \left(1 - \frac{1}{\epsilon_{r,m_\delta}} \right) D_{i,\delta}^{(m_\delta)} \\ D_{i,\text{Born}}^{(\text{bulk})} &= \frac{1}{d_i^{\text{Born}}} \\ D_{i,\text{SSM}}^{(\text{bulk})} &= \frac{1}{d_i^{\text{Born}} + \Delta d_i} - \frac{1}{d_i^{\text{Born}}} \\ D_{i,\text{DS}}^{(\text{ion})} &= \frac{1}{d_i^{\text{Born}}} - \frac{1}{d_i^{\text{Born}} + \Delta d_i} \end{aligned}$$



$$\tilde{\mu}_k^{\text{res}} = \tilde{a}^{\text{res}} + Z - 1 + \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial x_k} \right)_{T,v,x_i}$$

$$\ln \varphi_k = \tilde{\mu}_k^{\text{res}} - \ln Z, \quad a_i = \gamma_i x_i, \quad Q = \prod_i a_i^{\nu_i}$$

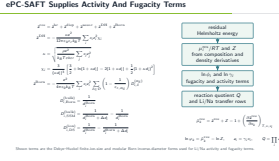
Shown terms are the Debye–Huckel finite-ion-size and modular Born inverse-diameter forms used for Li/Na activity and fugacity terms.

2026-05-15

Produced-Water Lithium Extraction └ Thermodynamic Extension

└ ePC-SAFT Supplies Activity And Fugacity Terms

This slide is the thermodynamic map in one place. ePC-SAFT starts from the residual Helmholtz energy and builds it from five physical contribution families: hard-chain packing and connectivity, dispersion attractions, association, Debye-Huckel electrolyte screening, and Born solvation or dielectric work. The Debye-Huckel term uses the screening parameter κ and finite-size correction χ_i from the documented equation set. The Born term is the modular form: it sums active Born, solvation shell, and dielectric-saturation inverse-diameter terms through $\mathcal{D}$. Once these terms are assembled, composition and density derivatives give residual chemical potentials and the compressibility factor. Those become fugacity coefficients and activity terms. For extraction, the important output is the activity basis for the reaction quotient and for the Li and Na transfer rows that the staged process model consumes.



Case-Study Coverage Matrix

Lane	Current basis	Process-facing use
feed basis	Smackover MS-2 and high-Li Smackover sensitivity	defines the main and feed-grade sensitivity cases
paper-response bridge	Rezaee Li/Na extraction responses with <i>O/A</i> conversion	supplies D_{Li} , D_{Na} , and $S_{Li/Na}$ for staged extraction
ALAMO-ready surrogate	current target-row fit for $\log D_{Li}$ and $\log D_{Na}$	provides the compact transfer surface used by the process model
PrOMMiS/IDAES stages	PrOMMiS/IDAES staged-extraction terms for Li and Na across 1, 3, and 5 stages	converts transfer variables to outlet recovery and raffinate behavior
screening TEA/UQ	stress, base, and favorable cases from the refreshed transfer surface	ranks operating levers for integration approval
direct ePC-SAFT transfer path	b4144c72 phase-tagged cross-phase route; 26 Rezaee rows evaluated	supplies the activity/fugacity layer for replacing the paper-response bridge

Process-facing rows remain stable as the thermodynamic basis is exchanged.

2026-05-15

Produced-Water Lithium Extraction └ Thermodynamic Extension

└ Case-Study Coverage Matrix

Case-Study Coverage Matrix		
Lane	Current basis	Process-facing use
feed basis	Smackover MS-2 and high-Li Smackover sensitivity	defines the main and feed-grade sensitivity cases
paper-response bridge	Rezaee Li/Na extraction responses with <i>O/A</i> conversion	supplies D_{Li} , D_{Na} , and $S_{Li/Na}$ for staged extraction
ALAMO-ready surrogate	current target-row fit for $\log D_{Li}$ and $\log D_{Na}$	provides the compact transfer surface used by the process model
PrOMMiS/IDAES stages	PrOMMiS/IDAES staged-extraction terms for Li and Na across 1, 3, and 5 stages	converts transfer variables to outlet recovery and raffinate behavior
screening TEA/UQ	stress, base, and favorable cases from the refreshed transfer surface	ranks operating levers for integration approval
direct ePC-SAFT transfer path	b4144c72 phase-tagged cross-phase route; 26 Rezaee rows evaluated	supplies the activity/fugacity layer for replacing the paper-response bridge
Process-facing rows remain stable as the thermodynamic basis is exchanged.		

This matrix is the scope control for the whole talk. The first five rows cover the feed basis, the paper-response bridge, the surrogate, the PrOMMiS/IDAES stages, and screening TEA. The last row is now concrete: the package commit used here evaluates 26 phase-tagged cross-phase Rezaee rows and supplies the activity and fugacity layer. The remaining integration task is to make that direct thermodynamic route feed the same process interface that currently receives the paper-response bridge. That is why the slide keeps the bridge and the direct route as separate rows.

The case study is a screening-quality bridge from Smackover feed chemistry to PrOMMiS/IDAES staged extraction and TEA, with ePC-SAFT activity and fugacity terms ready for direct process integration.

2026-05-15

Produced-Water Lithium Extraction
└─ Thermodynamic Extension

The case study is a screening-quality bridge from Smackover feed chemistry to PrOMMiS/IDAES staged extraction and TEA, with ePC-SAFT activity and fugacity terms ready for direct process integration.

The point of the package is now straightforward. The Smackover feed basis, solvent response, surrogate, staged extraction model, and screening TEA are already connected. The repaired ePC-SAFT route now gives the thermodynamic layer that this process chain needs: activity and fugacity terms that can be turned into direct Li and Na transfer rows. The practical next step is to make that route feed the same PrOMMiS and IDAES interface cleanly, then compare the staged recovery and TEA movement against the current paper-response bridge.

- 1. Smackover MS-2 is the main produced-water feed; the high-Li Smackover row defines the feed-grade sensitivity.
- 2. TBAC/DA DES + 10 wt% TOPO maps lithium and sodium transfer through ALAMO and into staged solvent extraction.
- 3. Three stages recover 47.85% lithium before the TEA pretreatment-loss adjustment.
- 4. The base screening TEA is 8.34 USD/kg LCE, with feed grade and phase ratio controlling the largest movement.
- 5. Ask: approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

2026-05-15

Produced-Water Lithium Extraction

- └ Thermodynamic Extension

- └ Technical Message

I would close on five points. First, Smackover MS-2 is the main feed basis, and high-lithium Smackover is the feed-grade sensitivity. Second, the active solvent basis is TBAC/DA DES with 10 weight percent TOPO, and that basis maps Li and Na transfer through ALAMO into staged extraction. Third, the three-stage case recovers 47.85 percent lithium before the TEA pretreatment adjustment. Fourth, the base screening TEA is 8.34 dollars per kilogram LCE, with feed grade and phase ratio driving the largest movement. Fifth, the decision request is to approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

Technical Message

- 1. Smackover MS-2 is the main produced-water feed; the high-Li Smackover row defines the feed-grade sensitivity.
- 2. TBAC/DA DES + 10 wt% TOPO maps lithium and sodium transfer through ALAMO and into staged solvent extraction.
- 3. Three stages recover 47.85% lithium before the TEA pretreatment-loss adjustment.
- 4. The base screening TEA is 8.34 USD/kg LCE, with feed grade and phase ratio controlling the largest movement.
- 5. Ask: approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

Supplemental Extension Context

Family	Why it belongs near produced water	Needed before modeling
CMM/REE ions	oilfield brines can carry critical-mineral value beyond Li	measured composition, aqueous speciation, and ion parameters
divalent ions	Ca, Mg, and Sr set pretreatment load and leakage risk	pretreatment data and extraction leakage bounds
hydrocarbons	oil-bearing water adds phase behavior and solvent-contact risks	oil-phase assay and phase-equilibrium data
ePC-SAFT extension	electrolyte and Born/dielectric terms give a thermodynamic route	accepted parameters and phase data for each species family

Boundary

Li/Na extraction remains the present case. CMM, REE, and hydrocarbon rows define where additional data would expand the same structure.

Use

The extension list guides what to measure before adding species, reactions, or oil-phase terms to the thermodynamic model.

USGS Energy Waste Science; critical-mineral produced-water review; electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature.

2026-05-15

Produced-Water Lithium Extraction

Thermodynamic Extension

Supplemental Extension Context

This supplemental slide is here to keep the extension story honest. Produced water is not only a lithium stream; it can contain critical-mineral ions, divalent ions, and hydrocarbon contact risks that matter for thermodynamics and operations. But none of those rows should be treated as solved by the current Li/Na case. CMM and REE rows need measured composition, speciation, and ion parameters. Divalent ions need pretreatment data and leakage bounds. Hydrocarbons need oil-phase assay and phase-equilibrium data. ePC-SAFT gives a credible thermodynamic route for extending the structure, but the data requirements have to come first.

Supplemental Extension Context		
Family	Why it belongs near produced water	Needed before modeling
CMM/REE ions	oilfield brines can carry critical-mineral value beyond Li	measured composition, aqueous speciation, and ion parameters
divalent ions	Ca, Mg, and Sr set pretreatment load and leakage risk	pretreatment data and extraction leakage bounds
hydrocarbons	oil-bearing water adds phase behavior and solvent-contact risks	oil-phase assay and phase-equilibrium data
ePC-SAFT extension	electrolyte and Born/dielectric terms give a thermodynamic route	accepted parameters and phase data for each species family
Boundary		Use
Li/Na extraction remains the present case. CMM, REE, and hydrocarbon rows define where additional data would expand the same structure.		The extension list guides what to measure before adding species, reactions, or oil-phase terms to the thermodynamic model.
USGS Energy Waste Science; critical-mineral produced-water review; electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature.		

- Rezaee et al. 2025: TBAC/DA DES + TOPO extraction response basis.
- Rezaee et al. 2026: PC-SAFT/ePC-SAFT parameter and reaction-equilibrium support.
- U.S. Geological Survey southern Arkansas Smackover brine data release, DOI: 10.5066/P9QPRYZN.
- USGS Energy Waste Science and produced-water lithium/mineral recovery context.
- Critical-mineral source potential review for U.S. oil and gas produced waters.
- Electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature for the extension basis.
- Produced-water feed table, surrogate fit table, PrOMMiS/IDAES staged-extraction table, and screening TEA table from the case-study result tables.

2026-05-15

Produced-Water Lithium Extraction
└ Thermodynamic Extension

└ Sources

The sources are grouped by role. Rezaee supplies the extraction response and the PC-SAFT/ePC-SAFT thermodynamic context. The USGS Smackover release supplies the produced-water feed basis, including the selected southern Arkansas row used in the main case. The USGS and produced-water mineral recovery literature explain why these brines are technically interesting beyond a single lithium number. The electrolyte PC-SAFT and ePC-SAFT Born/dielectric literature supports the thermodynamic extension path. The case-study tables are where the feed, surrogate, staged-extraction, and TEA values used in this presentation are collected.

Sources

• Rezaee et al. 2025: TBAC/DA DES + TOPO extraction response basis.
• Rezaee et al. 2026: PC-SAFT/ePC-SAFT parameter and reaction-equilibrium support.
• U.S. Geological Survey southern Arkansas Smackover brine data release, DOI: 10.5066/P9QPRYZN.
• USGS Energy Waste Science and produced-water lithium/mineral recovery context.
• Critical-mineral source potential review for U.S. oil and gas produced waters.
• Electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature for the extension basis.
• Produced-water feed table, surrogate fit table, PrOMMiS/IDAES staged-extraction table, and screening TEA table from the case-study result tables.